Synchronization in the Kuramoto Model

1 Introduction

In this model, each oscillator has its own natural frequency but interacts with all other oscillators through a coupling term. When the coupling strength is large enough, the oscillators tend to synchronize.

The objective of this simulation is to study the transition from incoherence to synchronization and characterize it using the **order parameter**.

2 Kuramoto Model

The dynamics of N coupled oscillators are described by

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) \tag{1}$$

where

- θ_i : phase of oscillator i
- ω_i : natural frequency of oscillator i
- K : coupling strength
- N : total number of oscillators

The first term represents the natural frequency of the oscillator, while the second term represents interaction with all other oscillators.

3 Order Parameter

Synchronization is quantified using the complex order parameter

$$z = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j} \tag{2}$$

The magnitude of this parameter gives the phase coherence

$$r = |z| \tag{3}$$

where

- $r = 0$: completely incoherent oscillators
- $r = 1$: perfectly synchronized oscillators

The phase of the order parameter is

$$\psi = \arg(z) \tag{4}$$

which represents the average phase of the oscillator population.

4 Simulation Method

The simulation steps were:

- Number of oscillators: $N = 200$
- Initial phases randomly distributed in $[0, 2\pi]$
- Natural frequencies drawn from a Gaussian distribution. We can also try Lorentzian distribution.
- Time step: $\Delta t = 0.05$
- Coupling strength K varied from 0 to 5

The oscillators were visualized on the unit circle using coordinates

$$x_i = \cos(\theta_i), \quad y_i = \sin(\theta_i)$$

Synchronization was studied by tracking the order parameter r over time.

5 Results

5.1 Oscillator Phases on Unit Circle

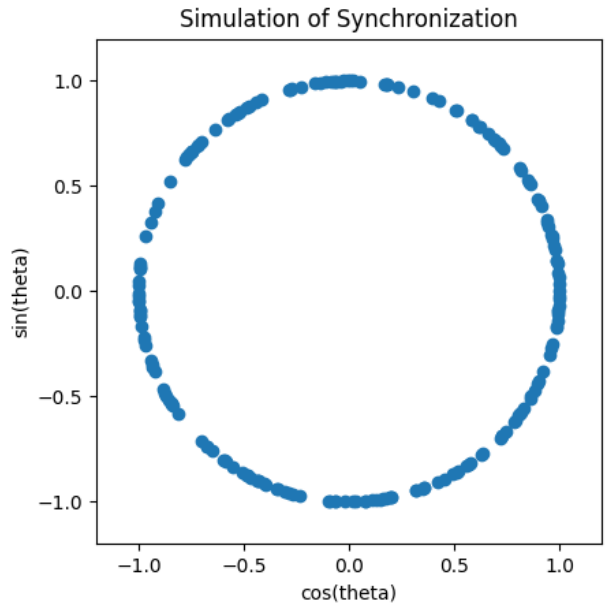

Figure 1: Positions of oscillators on the unit circle during the simulation (K=1).

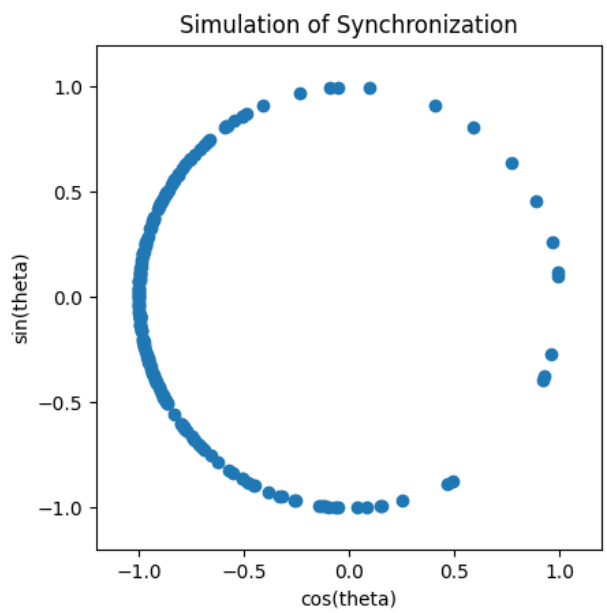

Figure 2: Positions of oscillators on the unit circle during the simulation (K=2, near Kc).

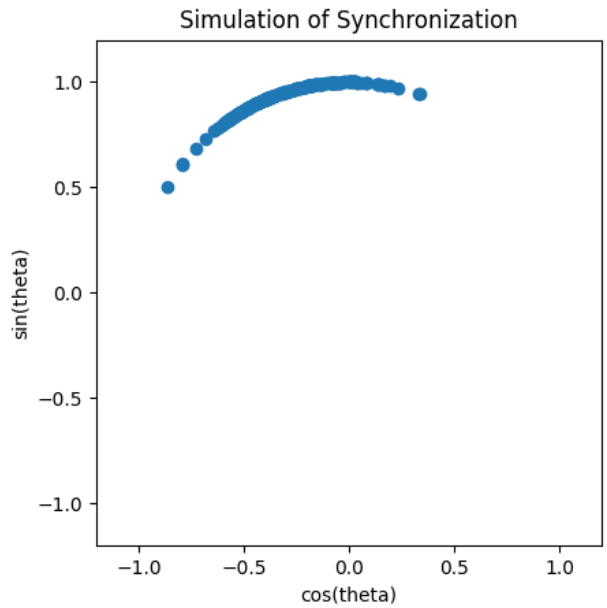

Figure 3: Positions of oscillators on the unit circle during the simulation (K=5).

5.2 Order Parameter vs Time

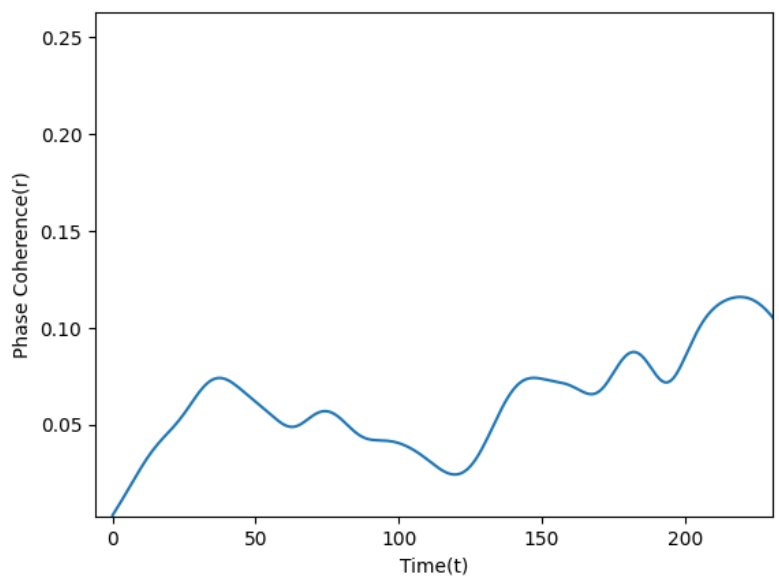

Figure 4: Evolution of phase coherence $r(t)$ with time (K=1).

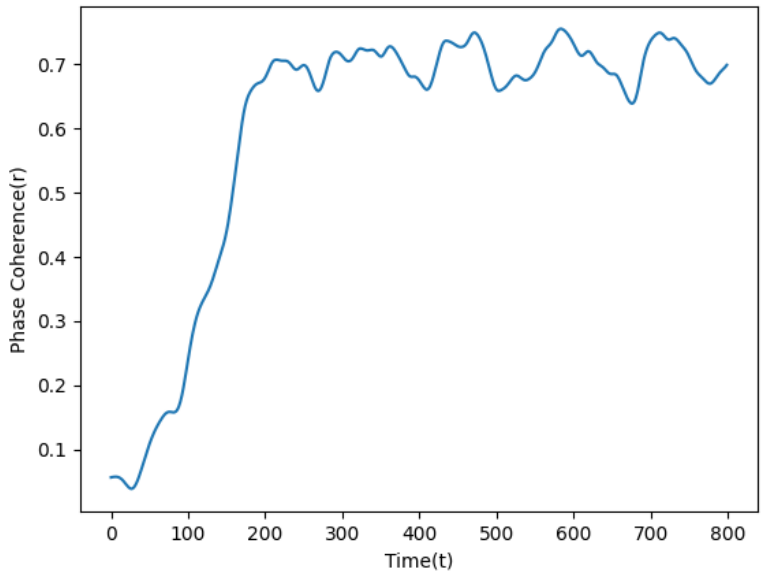

Figure 5: Evolution of phase coherence $r(t)$ with time (K=2, near Kc).

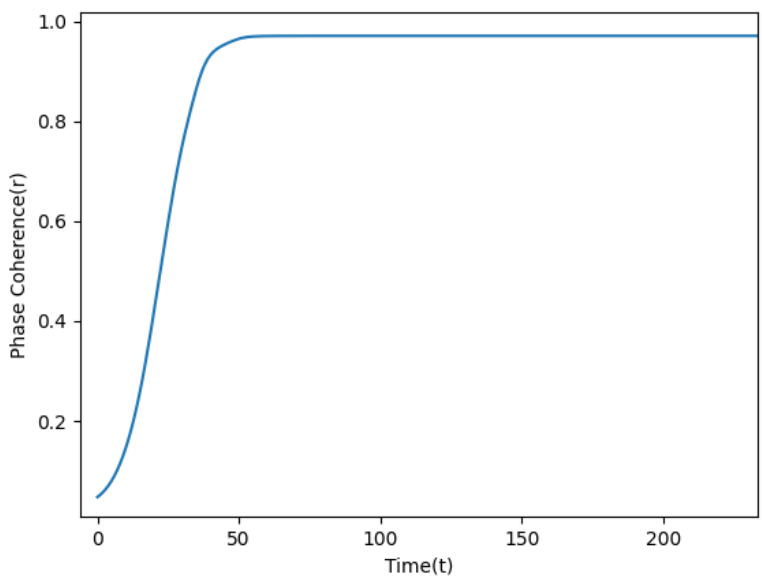

Figure 6: Evolution of phase coherence $r(t)$ with time (K=5).

5.3 Order Parameter vs Coupling Strength

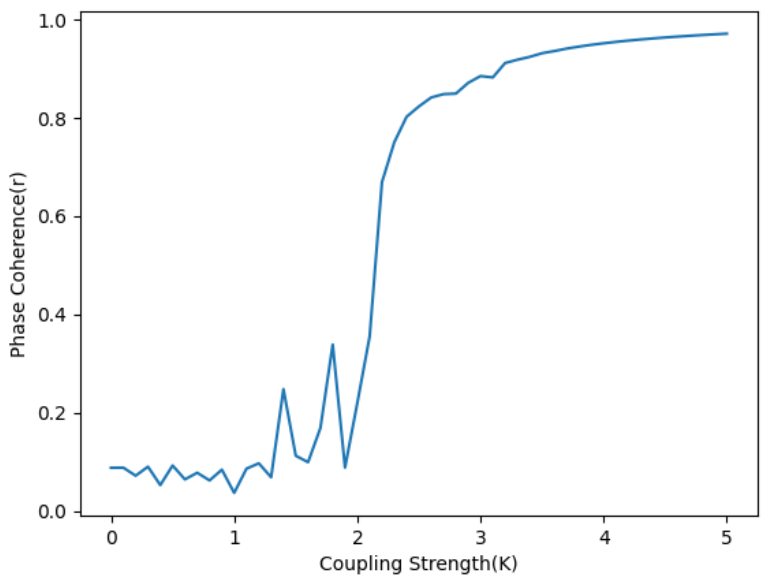

Figure 7: Dependence of synchronization measure r on coupling strength K.

6 Discussion

For small values of coupling strength K, the oscillators evolve almost independently and the order parameter remains close to zero, indicating incoherent behavior.

As K increases, the coupling between oscillators becomes stronger and synchronization begins to emerge. Above a certain threshold value of K, a significant fraction of oscillators lock to a common phase, leading to a rapid increase in the order parameter.

7 Conclusion

The simulation demonstrates the transition from incoherence to synchronization predicted by the Kuramoto model. The order parameter provides a simple quantitative measure of this transition.

This model illustrates how collective behavior can emerge from simple interactions between individual oscillators.